

Decision Trees Example Problem

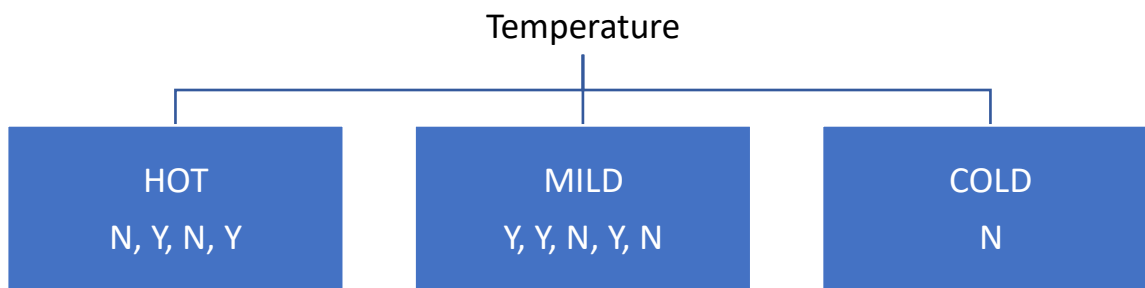
Consider the following data, where the Y label is whether or not the child goes out to play.

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Step 1: Calculate the IG (information gain) for each attribute (feature)

$$\begin{aligned}
 \text{Initial entropy} = H(Y) &= -\sum_y P(Y = y) \log_2 P(Y = y) \\
 &= -P(Y = \text{yes}) \log_2 P(Y = \text{yes}) - P(Y = \text{no}) \log_2 P(Y = \text{no}) \\
 &= -(0.5) \log_2(0.5) - (0.5) \log_2(0.5) \\
 &= 1
 \end{aligned}$$

Temperature:

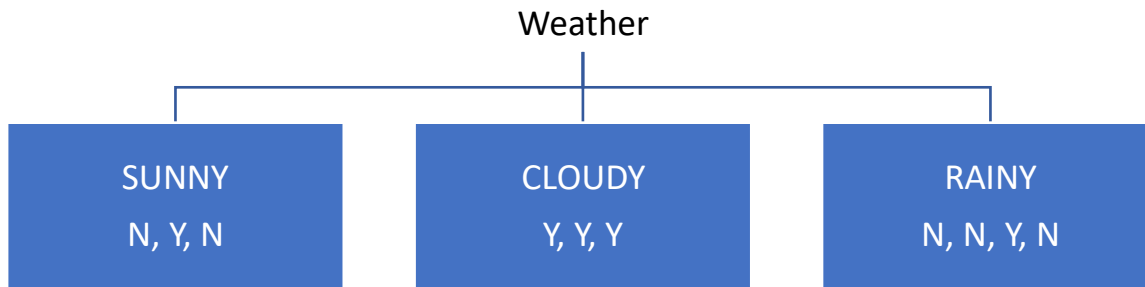


Total entropy of this division is:

$$\begin{aligned}
 H(Y | temp) &= -\sum_x P(temp = x) \sum_y P(Y = y | temp = x) \log_2 P(Y = y | temp = x) \\
 &= -(P(temp = H) \sum_y P(Y = y | temp = H) \log_2 P(Y = y | temp = H) + \\
 &\quad P(temp = M) \sum_y P(Y = y | temp = M) \log_2 P(Y = y | temp = M) + \\
 &\quad P(temp = C) \sum_y P(Y = y | temp = C) \log_2 P(Y = y | temp = C)) \\
 &= -((0.4)((\frac{1}{2}) \log_2 (\frac{1}{2}) + (\frac{1}{2}) \log_2 (\frac{1}{2})) + (0.5)((\frac{3}{5}) \log_2 (\frac{3}{5}) + (\frac{2}{5}) \log_2 (\frac{2}{5})) + \\
 &\quad (0.1)((1) \log_2 (1) + (0) \log_2 (0))) \\
 &= 0.7884
 \end{aligned}$$

$$IG(Y, temp) = 1 - 0.7884 = 0.2116$$

Weather:

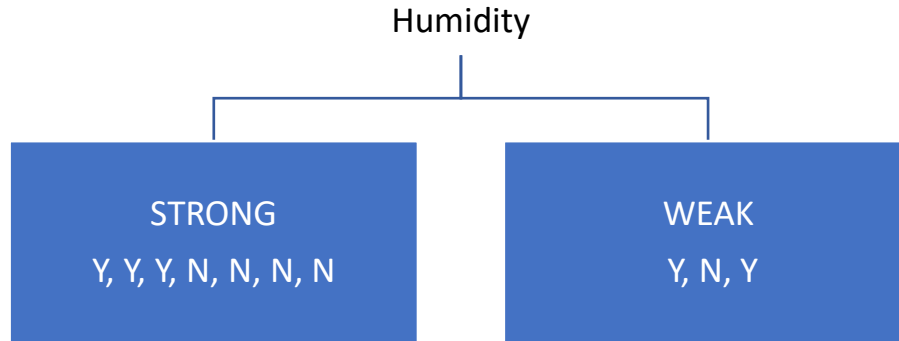


Total entropy of this division is:

$$\begin{aligned} H(Y | \text{weather}) &= - \sum_x P(\text{weather} = x) \sum_y P(Y = y | \text{weather} = x) \log_2 P(Y = y | \text{weather} = x) \\ &= -(P(\text{weather} = S) \sum_y P(Y = y | \text{weather} = S) \log_2 P(Y = y | \text{weather} = S) + \\ &\quad P(\text{weather} = C) \sum_y P(Y = y | \text{weather} = C) \log_2 P(Y = y | \text{weather} = C) + \\ &\quad P(\text{weather} = R) \sum_y P(Y = y | \text{weather} = R) \log_2 P(Y = y | \text{weather} = R)) \\ &= -((0.3)((\frac{1}{3}) \log_2 (\frac{1}{3}) + (\frac{2}{3}) \log_2 (\frac{2}{3})) + (0.3)((1) \log_2 (1) + (0) \log_2 (0)) + \\ &\quad (0.4)((\frac{1}{4}) \log_2 (\frac{1}{4}) + (\frac{3}{4}) \log_2 (\frac{3}{4}))) \\ &= 0.6 \end{aligned}$$

$$IG(Y, \text{weather}) = 1 - 0.6 = 0.4$$

Humidity:

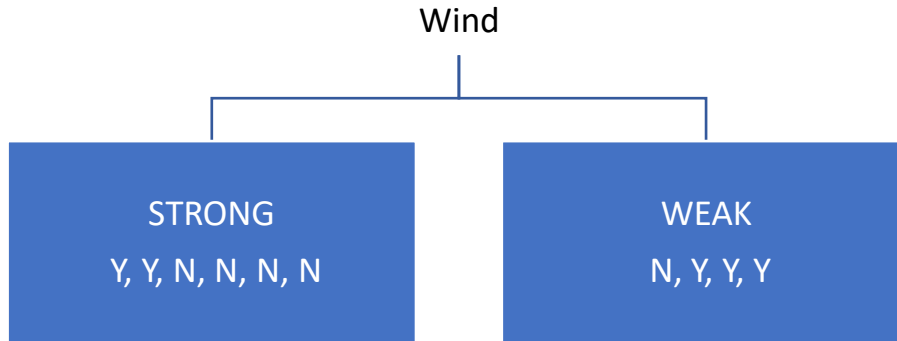


Total entropy of this division is:

$$\begin{aligned}
 H(Y | \text{hum}) &= - \sum_x P(\text{hum} = x) \sum_y P(Y = y | \text{hum} = x) \log_2 P(Y = y | \text{hum} = x) \\
 &= -(P(\text{hum} = H) \sum_y P(Y = y | \text{hum} = H) \log_2 P(Y = y | \text{hum} = H) + \\
 &\quad P(\text{hum} = N) \sum_y P(Y = y | \text{hum} = N) \log_2 P(Y = y | \text{hum} = N)) \\
 &= -((0.7)((\frac{3}{7}) \log_2 (\frac{3}{7}) + (\frac{4}{7}) \log_2 (\frac{4}{7})) + (0.3)((\frac{2}{3}) \log_2 (\frac{2}{3}) + (\frac{1}{3}) \log_2 (\frac{1}{3}))) \\
 &= 0.8651
 \end{aligned}$$

$$IG(Y, \text{hum}) = 1 - 0.8651 = 0.1349$$

Wind:



Total entropy of this division is:

$$\begin{aligned} H(Y | \text{wind}) &= - \sum_x P(\text{wind} = x) \sum_y P(Y = y | \text{wind} = x) \log_2 P(Y = y | \text{wind} = x) \\ &= -(P(\text{wind} = S) \sum_y P(Y = y | \text{wind} = S) \log_2 P(Y = y | \text{wind} = S) + \\ &\quad P(\text{wind} = W) \sum_y P(Y = y | \text{wind} = W) \log_2 P(Y = y | \text{wind} = W)) \\ &= -((0.6)((\frac{2}{6}) \log_2 (\frac{2}{6}) + (\frac{4}{6}) \log_2 (\frac{4}{6})) + (0.4)((\frac{1}{4}) \log_2 (\frac{1}{4}) + (\frac{3}{4}) \log_2 (\frac{3}{4}))) \\ &= 0.8755 \end{aligned}$$

$$IG(Y, \text{wind}) = 1 - 0.8755 = 0.1245$$

Step 2: Choose which feature to split with!

$$IG(Y, \text{wind}) = 0.1245$$

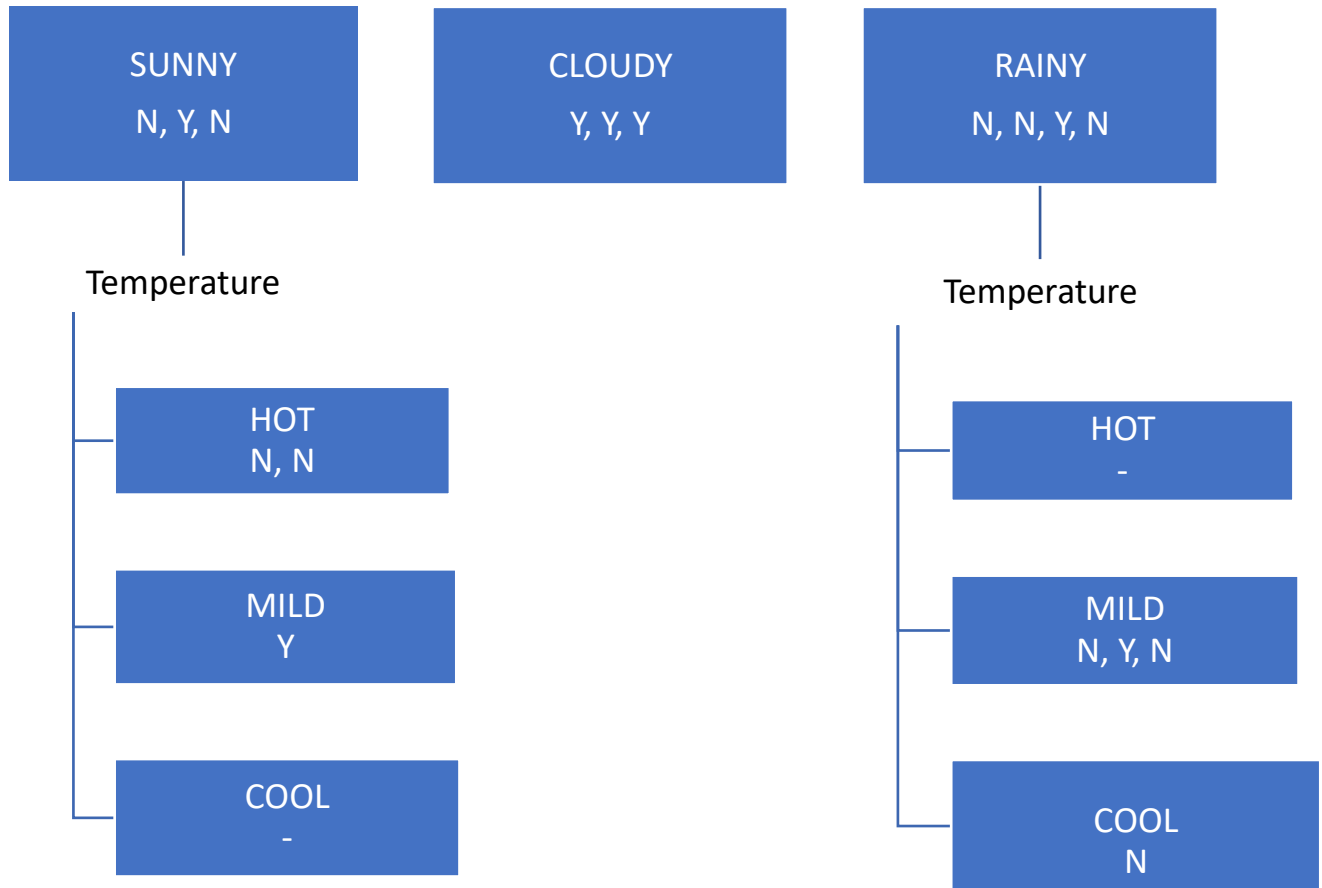
$$IG(Y, \text{hum}) = 0.1349$$

$$IG(Y, \text{weather}) = 0.4$$

$$IG(Y, \text{temp}) = 0.2116$$

Step 3: Repeat for each level (sad, I know)

Temperature



Entropy of "Sunny" node = $-\left(\left(\frac{1}{3}\right) \log_2 \left(\frac{1}{3}\right) + \left(\frac{2}{3}\right) \log_2 \left(\frac{2}{3}\right)\right) = 0.9183$

Entropy of its children = 0

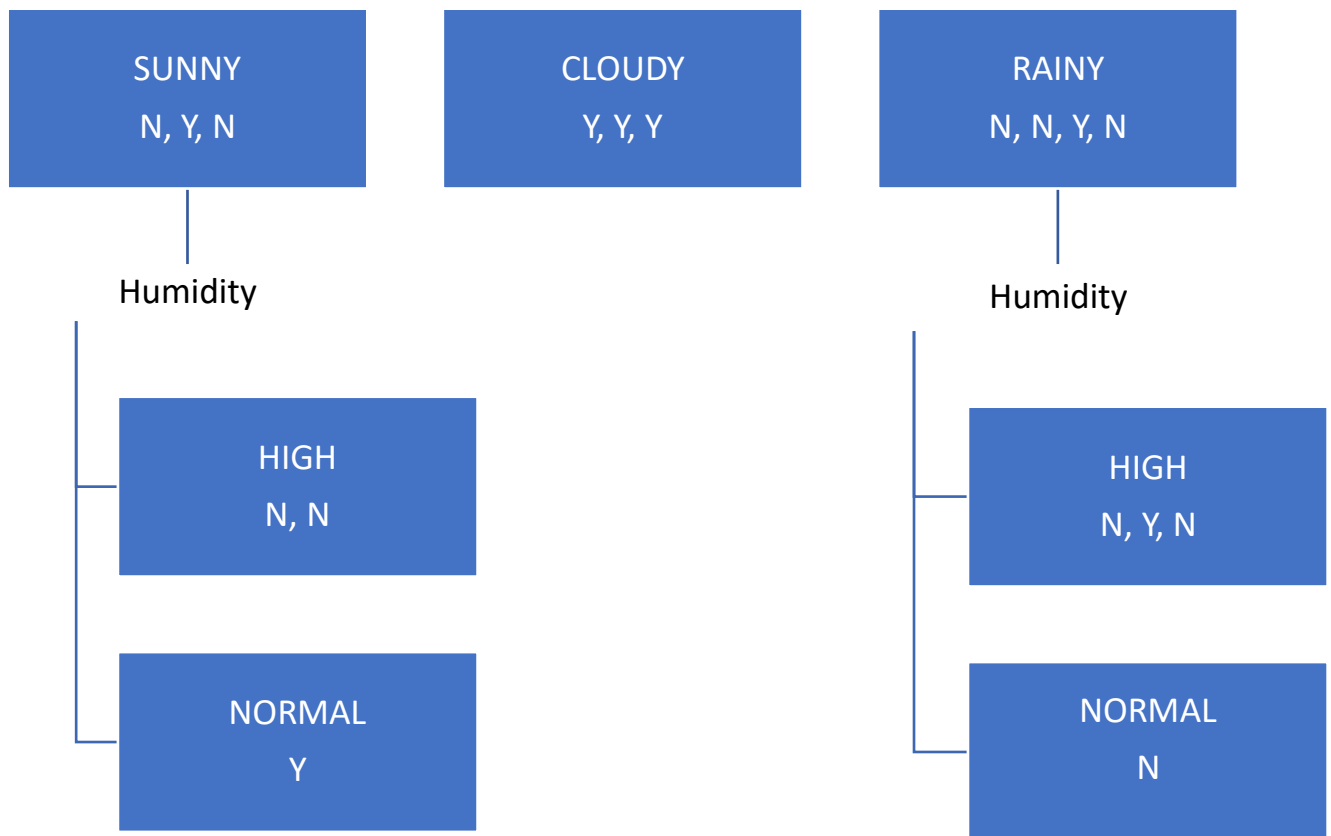
IG = 0.9183

Entropy of "Rainy" node = $-\left(\left(\frac{1}{4}\right) \log_2 \left(\frac{1}{4}\right) + \left(\frac{3}{4}\right) \log_2 \left(\frac{3}{4}\right)\right) = 0.8113$

Entropy of children = $-\left(\left(\frac{3}{4}\right) \left(\left(\frac{1}{3}\right) \log_2 \left(\frac{1}{3}\right) + \left(\frac{2}{3}\right) \log_2 \left(\frac{2}{3}\right)\right) + 0\right) = 0.6887$

IG = 0.1226

Humidity



Entropy of "Sunny" node = $-\left(\left(\frac{1}{3}\right) \log_2 \left(\frac{1}{3}\right) + \left(\frac{2}{3}\right) \log_2 \left(\frac{2}{3}\right)\right) = 0.9183$

Entropy of its children = 0

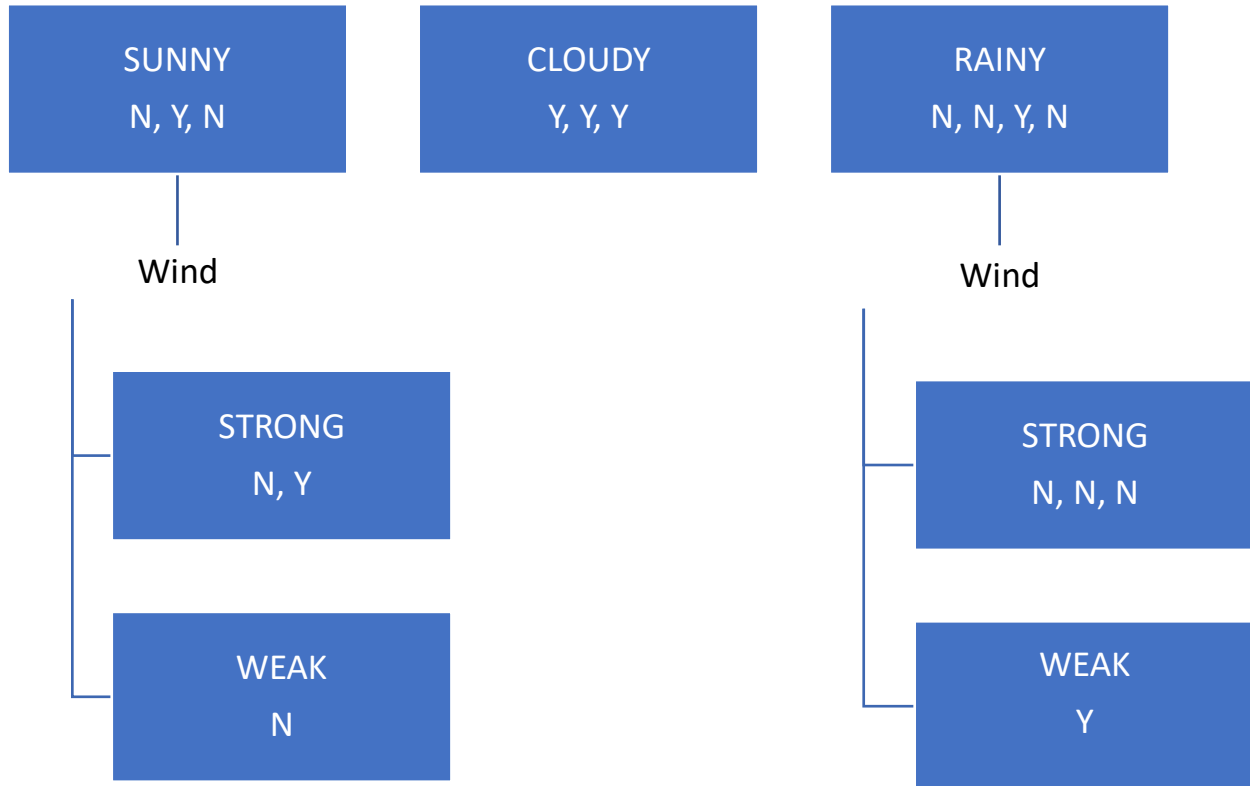
IG = 0.9183

Entropy of "Rainy" node = $-\left(\left(\frac{1}{4}\right) \log_2 \left(\frac{1}{4}\right) + \left(\frac{3}{4}\right) \log_2 \left(\frac{3}{4}\right)\right) = 0.8113$

Entropy of children = $-\left(\left(\frac{3}{4}\right) \left(\left(\frac{1}{3}\right) \log_2 \left(\frac{1}{3}\right) + \left(\frac{2}{3}\right) \log_2 \left(\frac{2}{3}\right)\right) + 0\right) = 0.6887$

IG = 0.1226

Wind



Entropy of "Sunny" node = $-\left(\left(\frac{1}{3}\right) \log_2 \left(\frac{1}{3}\right) + \left(\frac{2}{3}\right) \log_2 \left(\frac{2}{3}\right)\right) = 0.9183$

Entropy of its children = $-\left(\frac{2}{3}\right)\left(\left(\frac{1}{2}\right) \log_2 \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) \log_2 \left(\frac{1}{2}\right)\right) + 0 = 0.6667$

IG = 0.2516

Entropy of "Rainy" node = $-\left(\left(\frac{1}{4}\right) \log_2 \left(\frac{1}{4}\right) + \left(\frac{3}{4}\right) \log_2 \left(\frac{3}{4}\right)\right) = 0.8113$

Entropy of children = 0

IG = 0.8113

Step 4: Choose feature for each node to split on!

“Sunny node”:

$$IG(Y, \text{weather}) = IG(\text{humidity}) = 0.9183$$

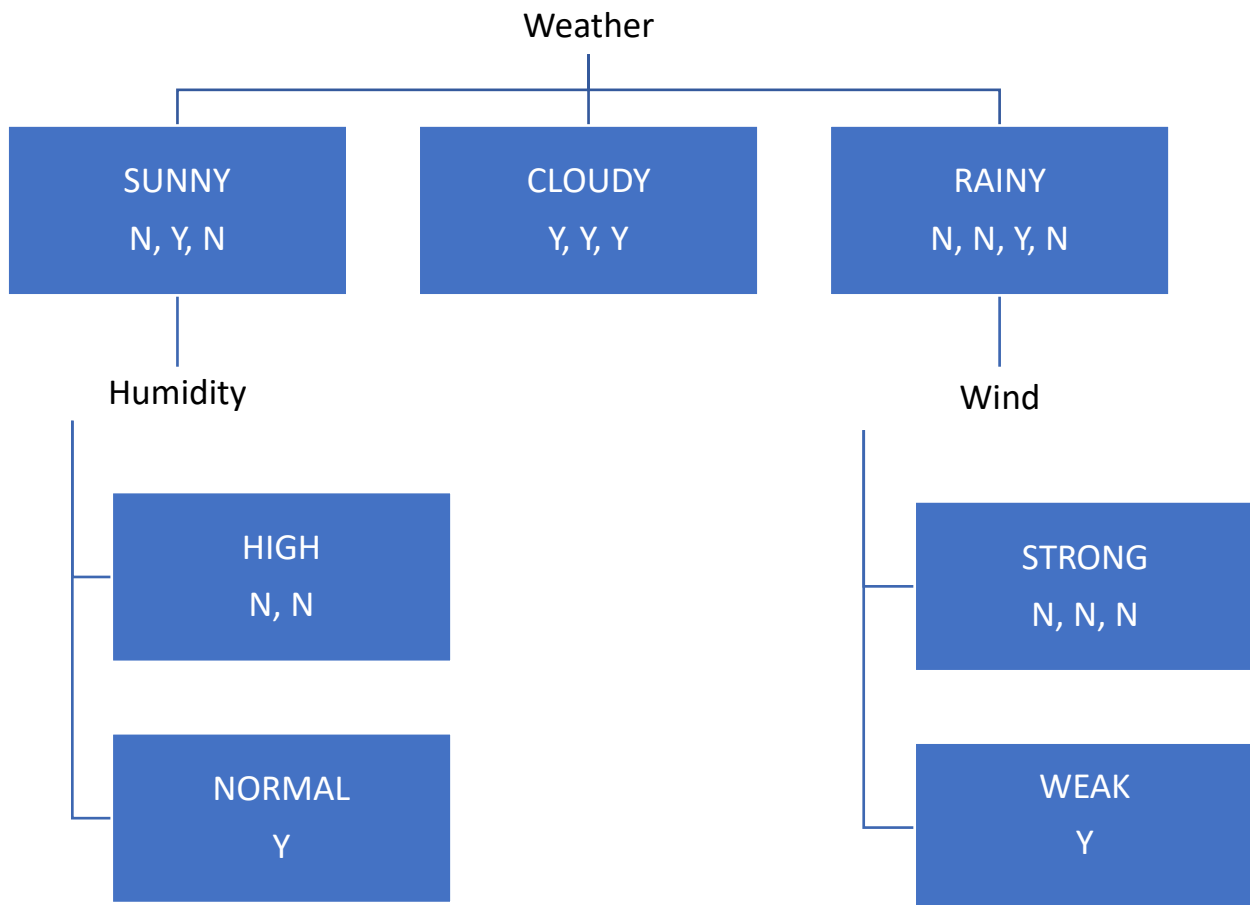
$$IG(Y, \text{wind}) = 0.2516$$

“Rainy node”:

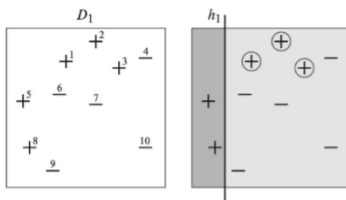
$$IG(Y, \text{weather}) = IG(Y, \text{humidity}) = 0.1226$$

$$IG(Y, \text{wind}) = 0.8113$$

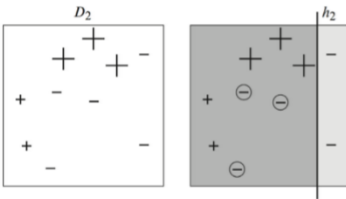
Final Tree!



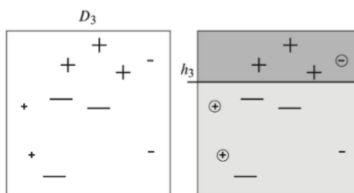
Boosting



	1	2	3	4	5	6	7	8	9	10	
$D_1(i)$	<u>0.10</u>	<u>0.10</u>	<u>0.10</u>	0.10	0.10	0.10	0.10	0.10	0.10	0.10	$\epsilon_1 = 0.30, \alpha_1 \approx 0.42$
$e^{-\alpha_1 y_i h_1(x_i)}$	1.53	1.53	1.53	0.65	0.65	0.65	0.65	0.65	0.65	0.65	
$D_1(i) e^{-\alpha_1 y_i h_1(x_i)}$	0.15	0.15	0.15	0.07	0.07	0.07	0.07	0.07	0.07	0.07	$Z_1 \approx 0.92$



$D_2(i)$	0.17	0.17	0.17	0.07	0.07	<u>0.07</u>	<u>0.07</u>	0.07	<u>0.07</u>	0.07	$\epsilon_2 \approx 0.21, \alpha_2 \approx 0.65$
$e^{-\alpha_2 y_i h_2(x_i)}$	0.52	0.52	0.52	0.52	0.52	1.91	1.91	0.52	1.91	0.52	
$D_2(i) e^{-\alpha_2 y_i h_2(x_i)}$	0.09	0.09	0.09	0.04	0.04	0.14	0.14	0.04	0.14	0.04	$Z_2 \approx 0.82$



$D_3(i)$	0.11	0.11	0.11	<u>0.05</u>	<u>0.05</u>	0.17	0.17	<u>0.05</u>	0.17	0.05	$\epsilon_3 \approx 0.14, \alpha_3 \approx 0.92$
$e^{-\alpha_3 y_i h_3(x_i)}$	0.40	0.40	0.40	2.52	2.52	0.40	0.40	2.52	0.40	0.40	
$D_3(i) e^{-\alpha_3 y_i h_3(x_i)}$	0.04	0.04	0.04	0.11	0.11	0.07	0.07	0.11	0.07	0.02	$Z_3 \approx 0.69$

$$H = \text{sign} \left(0.42 \begin{array}{|c|} \hline \text{gray} \\ \hline \end{array} + 0.65 \begin{array}{|c|} \hline \text{gray} \\ \hline \end{array} + 0.92 \begin{array}{|c|} \hline \text{gray} \\ \hline \end{array} \right) = \begin{array}{|c|} \hline \text{gray} \\ \hline \end{array}$$

(https://www.ccs.neu.edu/home/vip/teach/MLcourse/4_boosting/slides/boosting.pdf)